

IT SUPPORT HOMELAB PROJECT

Windows IT Support & Troubleshooting

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Project Type: Hands-on IT Support Homelab

Operating System: Windows 10/11

Project Overview

This project demonstrates practical IT Support and Windows troubleshooting skills completed on a personal computer.

The project covers:

- Windows Task Manager
- Windows Services
- Network Troubleshooting
- System File Checking
- Disk and Storage Checking
- Windows Event Viewer

The project includes screenshots and documentation as evidence of the practical tasks completed.

Task 1: Windows Task Manager

Objective

To use Windows Task Manager to monitor system performance and identify applications and processes currently running on the computer.

Tools Used

Windows Task Manager

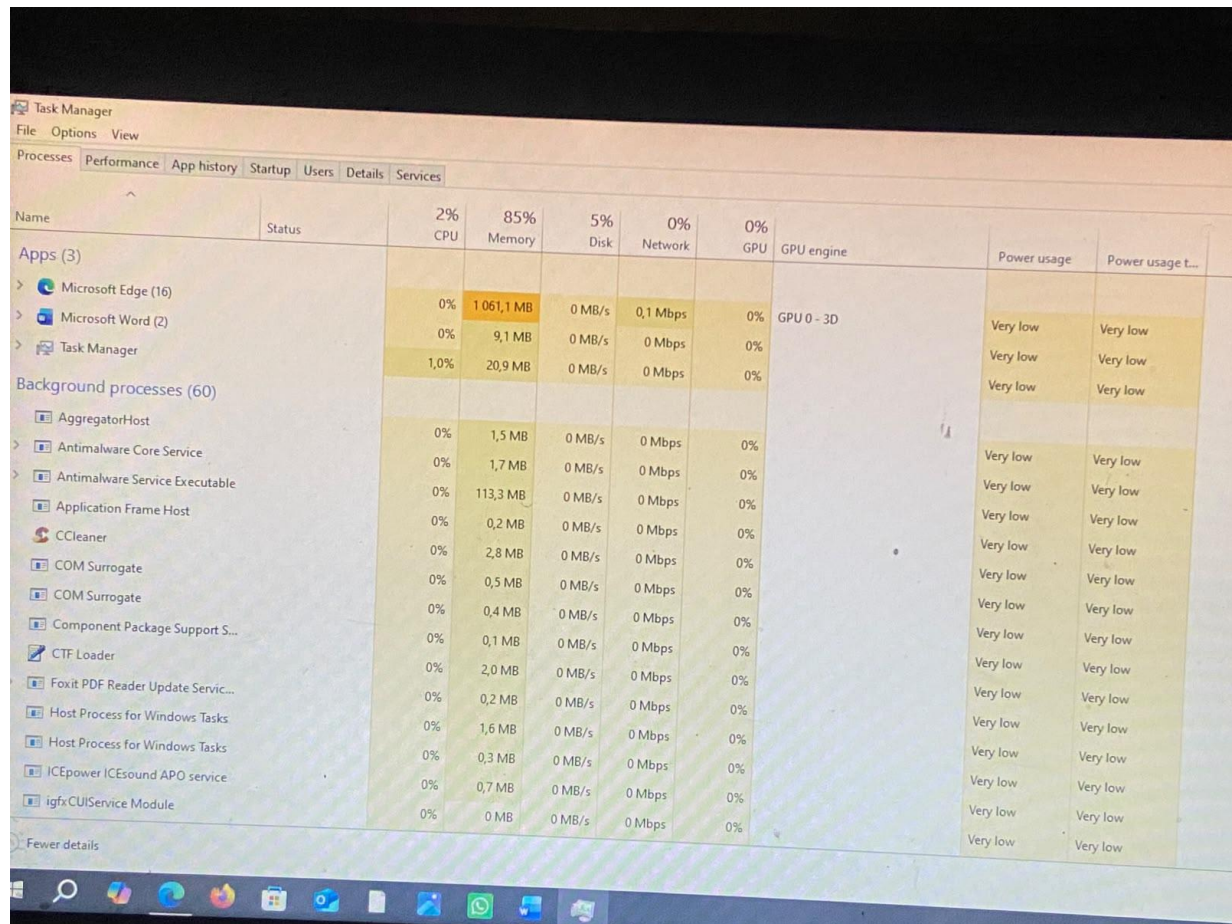
Windows 10/11

Procedure

1. I opened Windows Task Manager using Ctrl + Shift + Esc.
2. I reviewed the Processes tab to identify running applications and background processes.
3. I checked the CPU, Memory, Disk and Network usage.
4. I opened the Performance tab to monitor the computer's system resources.
5. I reviewed the Startup apps section to identify applications configured to start with Windows.

Evidence

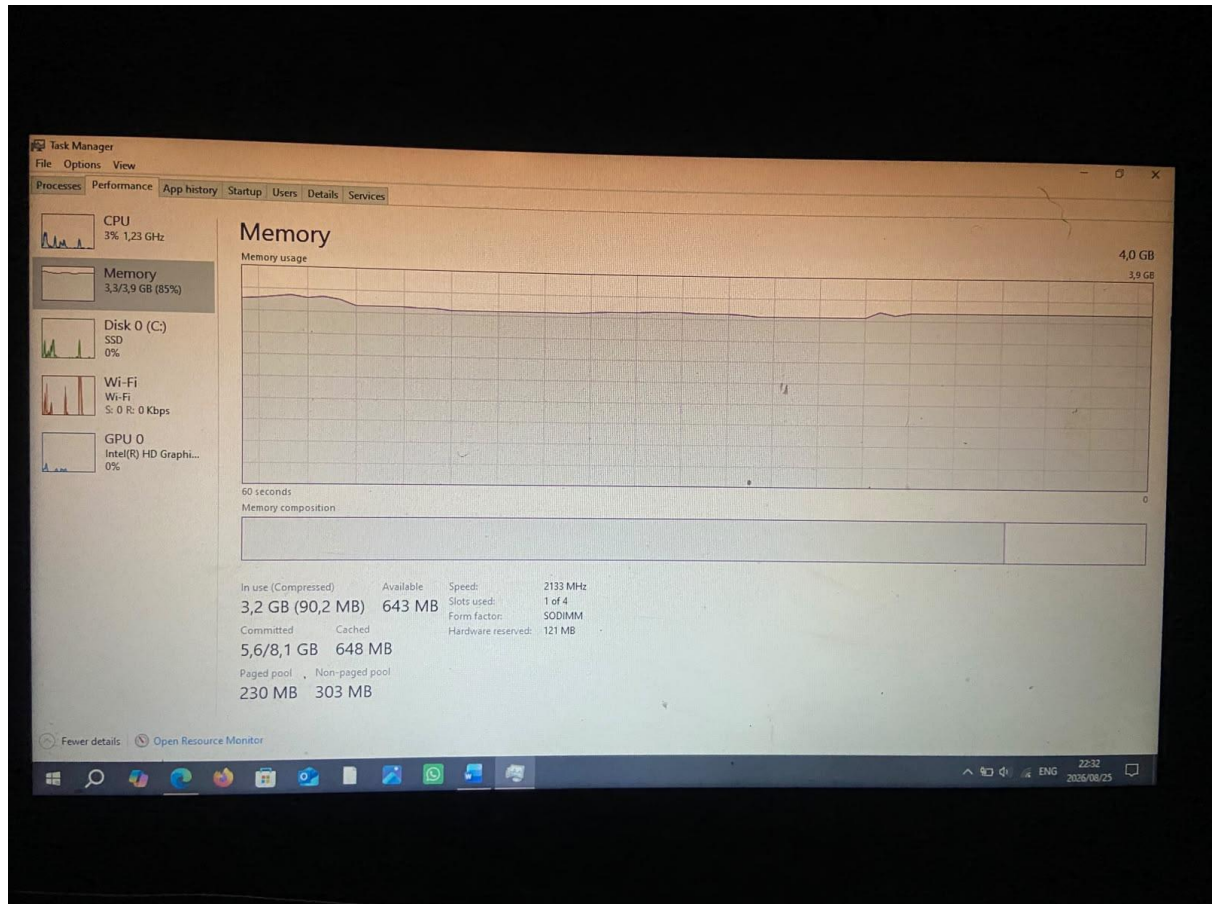
Screenshot 1 -Processes



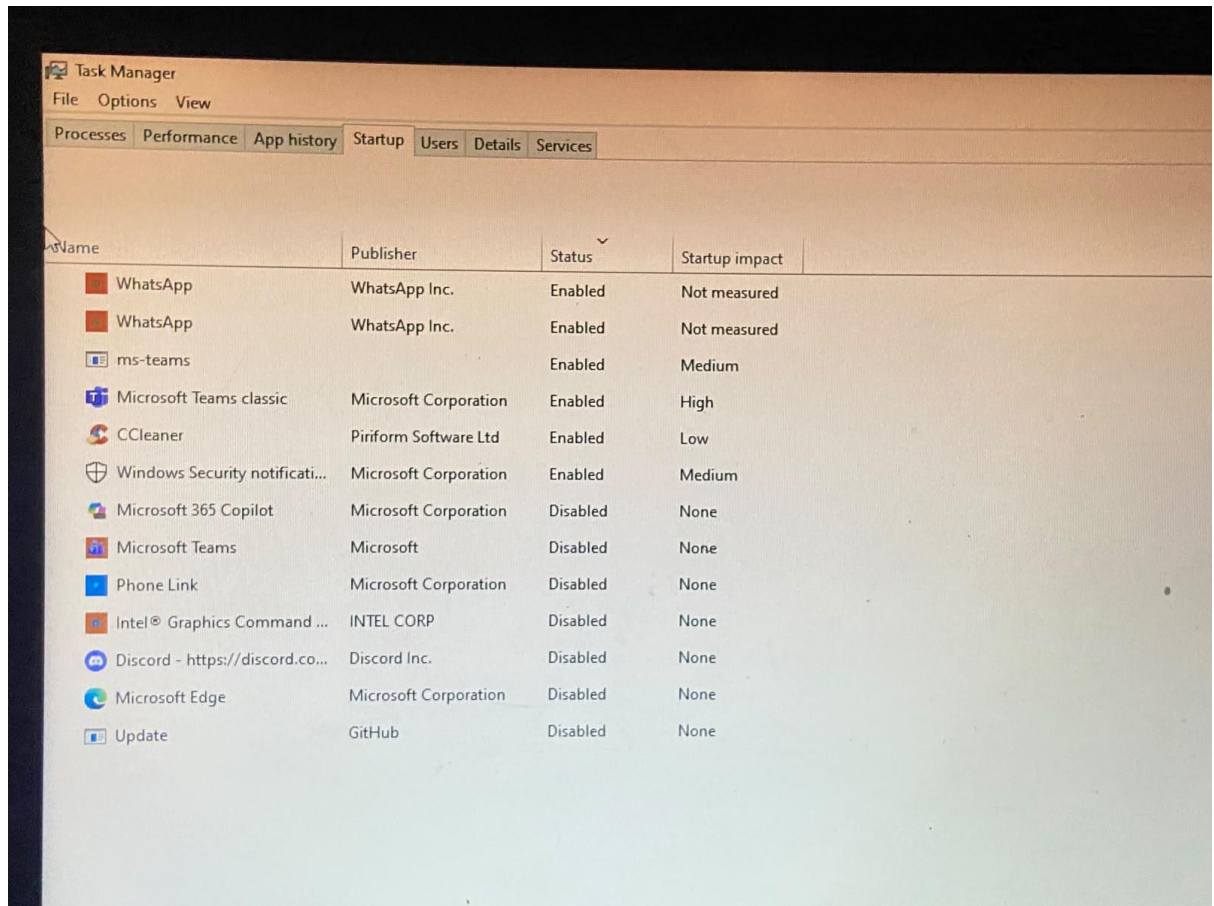
The screenshot shows the Windows Task Manager Performance tab. At the top, a summary bar displays overall system usage: 29% CPU, 85% Memory, 5% Disk, 0% Network, and 0% GPU. Below this, the 'Processes' tab is active, showing a list of running applications and background processes. The list is organized into sections: 'Apps (3)' and 'Background processes (60)'. Each process entry includes its name, status, and a row of resource usage metrics (CPU, Memory, Disk, Network, GPU) followed by power usage information. The 'Microsoft Edge' process is highlighted in yellow, showing 0% CPU, 1,061.1 MB Memory, 0 MB/s Disk, 0.1 Mbps Network, and 0% GPU usage. The 'Background processes' section lists various system services and background applications, all showing low resource usage. The taskbar at the bottom shows several open applications, including the Start menu, Search, File Explorer, and various utility apps.

Name	Status	29% CPU	85% Memory	5% Disk	0% Network	0% GPU	GPU engine	Power usage	Power usage t...
Apps (3)									
Microsoft Edge (16)		0%	1,061.1 MB	0 MB/s	0.1 Mbps	0%	GPU 0 - 3D	Very low	Very low
Microsoft Word (2)		0%	9.1 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Task Manager		1.0%	20.9 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Background processes (60)									
AggregatorHost		0%	1.5 MB	0 MB/s	0 Mbps	0%			
Antimalware Core Service		0%	1.7 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Antimalware Service Executable		0%	113.3 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Application Frame Host		0%	0.2 MB	0 MB/s	0 Mbps	0%		Very low	Very low
CCleaner		0%	2.8 MB	0 MB/s	0 Mbps	0%		Very low	Very low
COM Surrogate		0%	0.5 MB	0 MB/s	0 Mbps	0%		Very low	Very low
COM Surrogate		0%	0.4 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Component Package Support S...		0%	0.1 MB	0 MB/s	0 Mbps	0%		Very low	Very low
CTF Loader		0%	2.0 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Foxit PDF Reader Update Servic...		0%	0.2 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Host Process for Windows Tasks		0%	1.6 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Host Process for Windows Tasks		0%	0.3 MB	0 MB/s	0 Mbps	0%		Very low	Very low
ICEpower ICESound APO service		0%	0.7 MB	0 MB/s	0 Mbps	0%		Very low	Very low
igfxCUIService Module		0%	0 MB	0 MB/s	0 Mbps	0%		Very low	Very low

Screenshot 2 -Performance



Screenshot 3 -Startup App



Result

Task Manager successfully displayed the computer's running processes, resource usage and startup applications.

- Skills Demonstrated
- Windows troubleshooting
- System performance monitoring
- Task Manager
- Process management
- Resource monitoring
- Technical documentation

Task 2: Windows Services

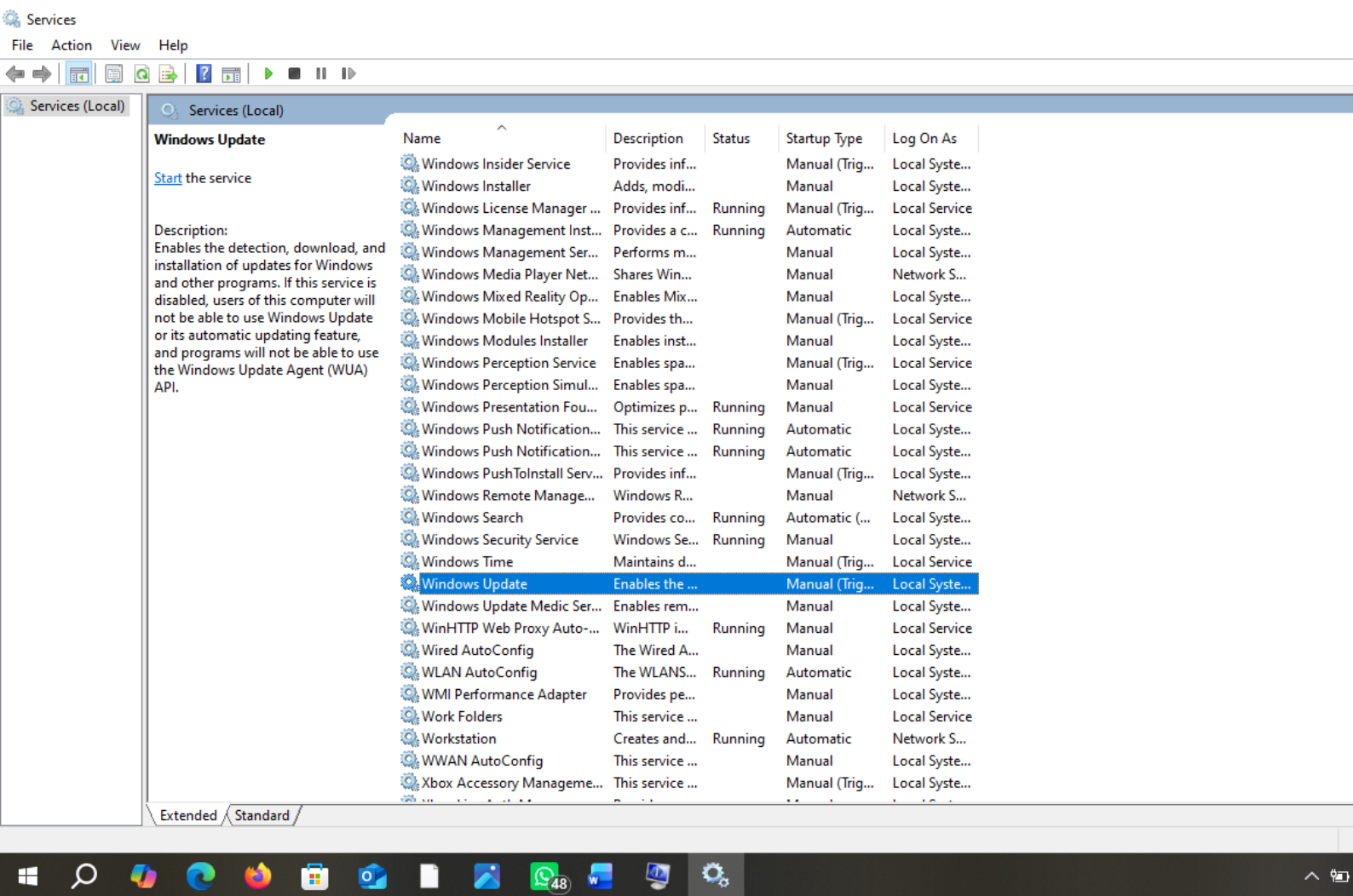
Objective

To investigate Windows services and understand how IT Support technicians can monitor services that run in the background.

Tools Used

- Windows Services (services.msc)
- Procedure
- I opened the Windows Run dialog using Windows Key + R.
- I entered services. MSc and opened the Windows Services console.
- I located the Windows Update service.
- I reviewed the service's status, startup type and description.

Evidence Screenshot 1 – Windows Update Service



Result

The screenshot shows the Windows Services console with the Windows Update service selected. I reviewed its service status, startup type, and description without changing the system configuration.

- Skills Demonstrated
- Windows Services management
- System administration

- Windows troubleshooting
- Technical documentation

Task 3: Network Troubleshooting

Objective

To check the computer's network configuration and test network connectivity using Command Prompt.

Tools used

- Windows Command Prompt
- ipconfig
- ping

Procedure

Step 1 – Open Command Prompt

I opened Command Prompt on the computer to perform basic network troubleshooting.

Step 2 – Check Network Configuration

I entered the ipconfig command to view the computer's network configuration.

The command displayed information including the IPv4 address, subnet mask and default gateway for the active Wi-Fi connection.

Evidence

Screenshot 1- ipconfig Network Configuration

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ > ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::bf01:9517:758b:801d%8
    IPv4 Address. . . . . : 192.168.18.85
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.18.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\Users\ >
```

Step 3 -Test Network Connectivity

I entered the command:

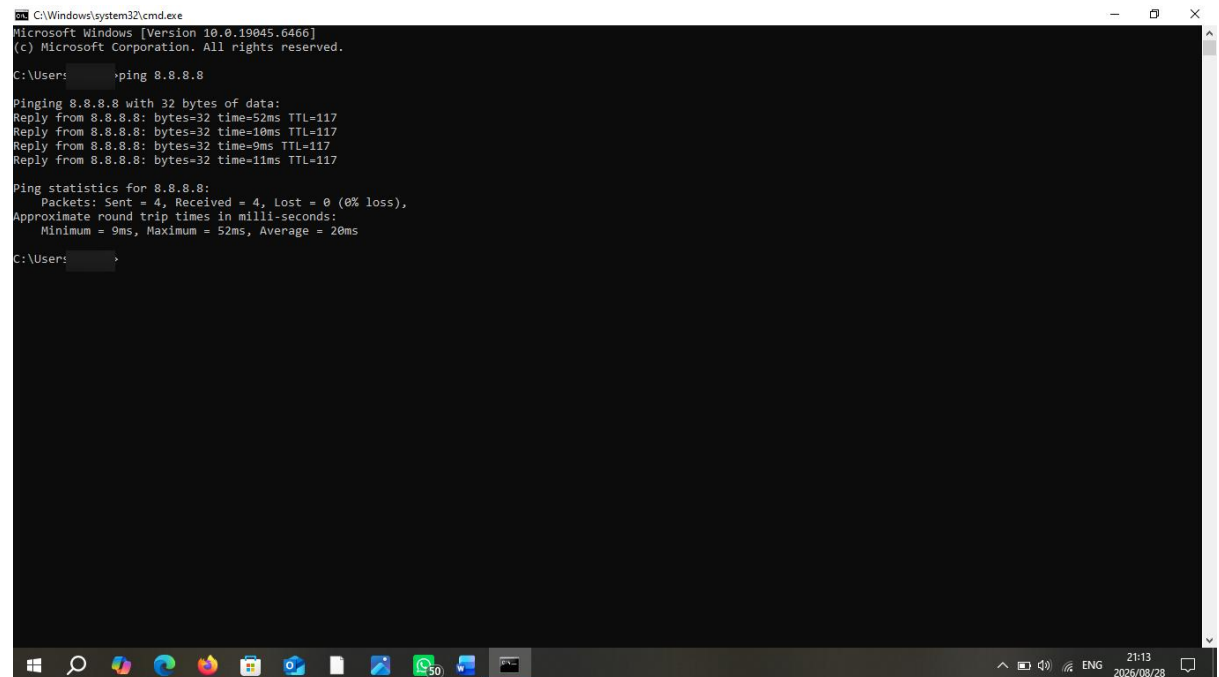
ping 8.8.8.8

The computer sent four packets and successfully received all four packets, resulting in 0% packet loss.

This confirmed successful communication with the tested address.

Evidence

Screenshot 2 -Ping Connectivity Test

A screenshot of a Windows Command Prompt window. The title bar reads 'C:\Windows\system32\cmd.exe'. The window content shows the output of a 'ping 8.8.8.8' command. It displays four successful replies from 8.8.8.8 with varying times and TTL values. Below the replies, it shows ping statistics: 4 packets sent, 4 received, 0% loss, and approximate round trip times (9ms, 52ms, 11ms, 117ms). The command prompt is at 'C:\User\>'. The Windows taskbar is visible at the bottom with various icons and a system clock showing 21:13 on 2026/08/28.

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\User>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=52ms TTL=117
Reply from 8.8.8.8: bytes=32 time=10ms TTL=117
Reply from 8.8.8.8: bytes=32 time=9ms TTL=117
Reply from 8.8.8.8: bytes=32 time=11ms TTL=117

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 52ms, Average = 20ms

C:\User>
```

ping 8.8.8.8

Result

I successfully checked the computer's network configuration using ipconfig and tested network connectivity using ping. The connectivity test showed successful communication with 0% packet loss.

Skills Demonstrated

- Network troubleshooting
- Using Command Prompt

- IP configuration
- Connectivity testing
- Basic network diagnostics

Technical documentation

Task 4: System File Check

Objective:

To check Windows system files for corruption and repair any issues found.

Tool Used:

Windows System File Checker (SFC)

Procedure:

1. I opened Command Prompt as an Administrator.
2. I ran the command `sfc /scannow`.
3. Windows scanned the protected system files.
4. The scan reached 100% completion.
5. Windows detected corrupt system files and successfully repaired them.

Evidence:

Screenshot 1 - System File Checker Scan

```
Administrator: Command Prompt
[/OFFWINDIR=<offline windows directory> /OFFBOOTDIR=<offline boot directory> [/OFFLOGFILE=<log file path>]

/SCANNOW      Scans integrity of all protected system files and repairs files with
               problems when possible.
/VERIFYONLY   Scans integrity of all protected system files. No repair operation is
               performed.
/SCANFILE     Scans integrity of the referenced file, repairs file if problems are
               identified. Specify full path <file>
/VERIFYFILE   Verifies the integrity of the file with full path <file>. No repair
               operation is performed.
/OFFBOOTDIR   For sfc /scannow r, specify the location of the offline boot directory
/OFFWINDIR     For offline repair, specify the location of the offline windows directory
Beginning system scan. This process will take some time.
Beginning verification phase of system scan.
Verification 100% complete.
sfc /SCANNOW
Windows Resource Protection found corrupt files and successfully repaired them.
For online repairs, details are included in the CBS log file located at %windir%\
windir\Logs\CBS\CBS.log. For example C:\Windows\Logs\CBS\CBS.log. For offlineNDIR=d:\windows /OFFLOGFILE=c:\
repairs, details are included in the log file provided by the /OFFLOGFILE flag.

C:\Windows\system32>sfc /scannow
```

Result:

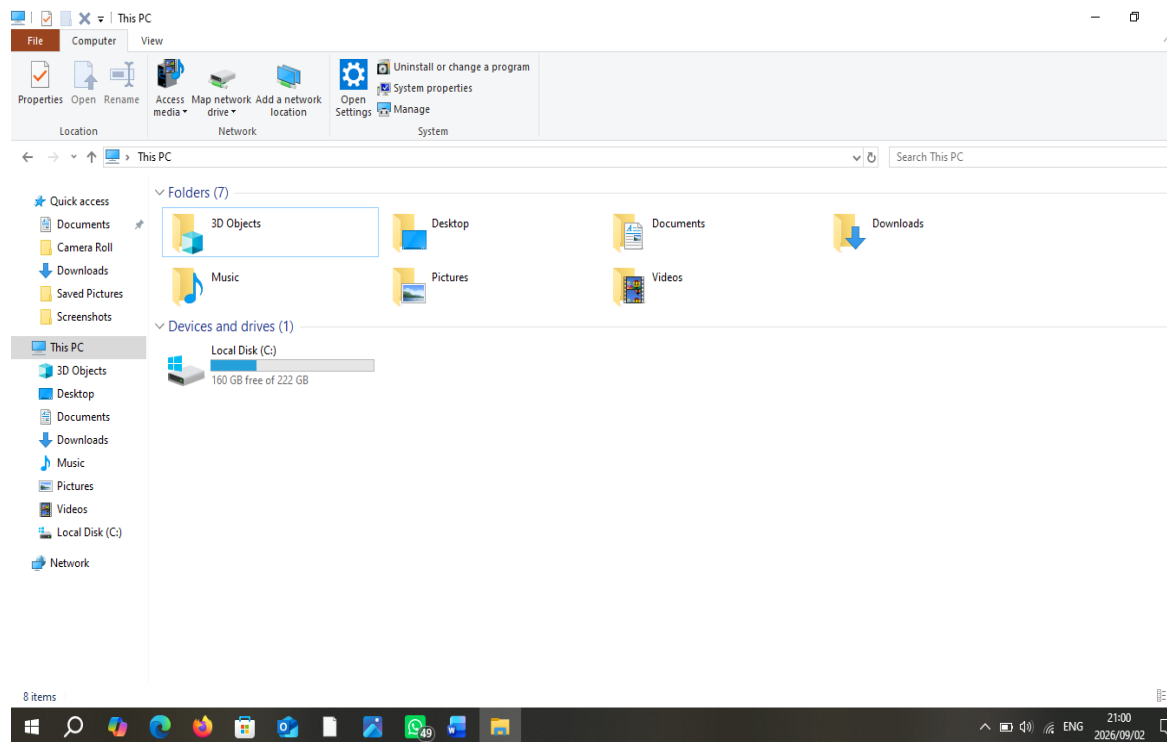
- Windows Resource Protection found corrupt files and successfully repaired them.
- Skills Demonstrated:
- Windows troubleshooting
- Command Prompt
- System file troubleshooting
- Basic system repair

Technical documentation

Task 5: Disk and Storage Check

Objective:

To check the available storage space on the computer.



Result:

I checked the Local Disk (C:) to review the computer's storage capacity. The computer had approximately 160 GB of free space out of a total capacity of 222 GB.

Skills Demonstrated:

- Storage management
- Basic Windows troubleshooting
- System monitoring
- Technical documentation

Task 6: Windows Event Viewer

Objective:

To use Windows Event Viewer to identify and investigate system warnings.

Tool Used:

Windows Event Viewer

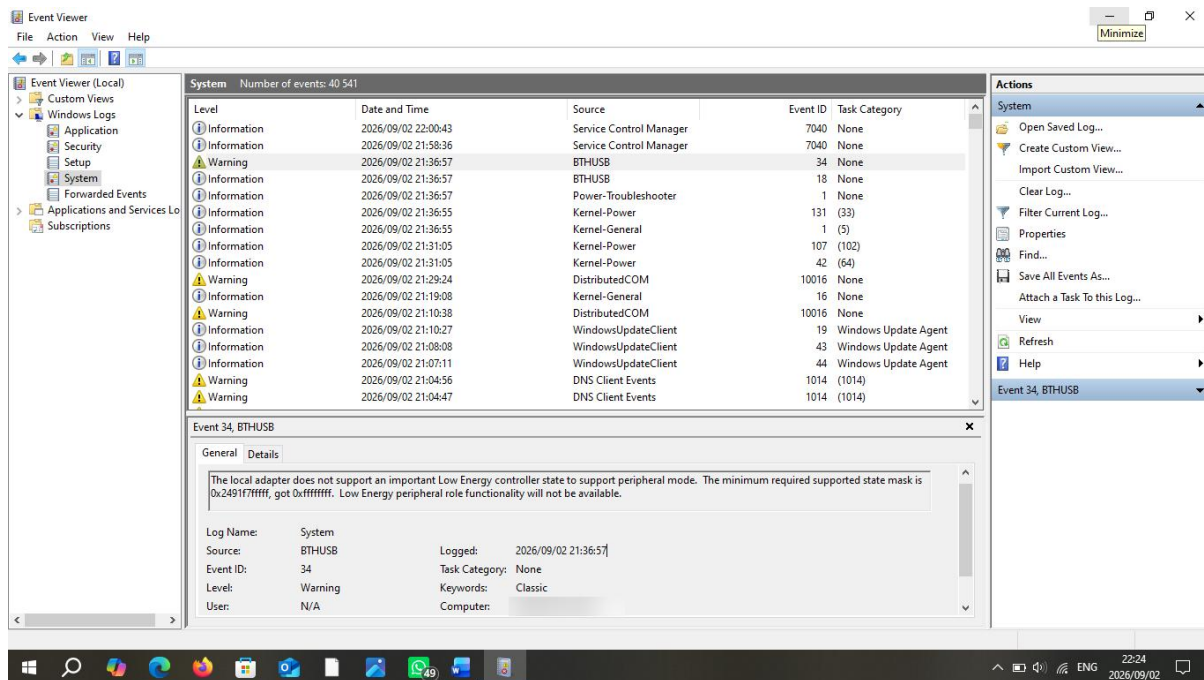
1. I opened Event Viewer using eventvwr.msc.
2. I opened Windows Logs → System.
3. I reviewed the events listed in the System log.
4. I selected a warning event to investigate.
5. I recorded the event source, Event ID and warning level.

Event Details:

- Source: BTUSB
- Event ID: 34
- Level: Warning

Evidence:

Screenshot 1 - BTUSB Event Viewer Warning



Investigation:

The selected event was recorded as a Warning generated by the BTHUSB source. I reviewed the event information to practise identifying system warnings that could be investigated during an IT Support troubleshooting process. The Event ID and source were recorded so that the issue could be researched further if it affected the computer.

Result:

I successfully used Windows Event Viewer to identify and review a system warning. I recorded the event source, Event ID, and warning level as part of the troubleshooting process.

Skills Demonstrated:

- Event Viewer
- Basic troubleshooting
- Investigating system events
- Windows system monitoring
- Technical documentation

Project Conclusion

This hands-on IT Support Home lab Project allowed me to practice practical Windows troubleshooting and IT Support skills on a personal computer. I worked with Task Manager, Windows Services, Command Prompt, network troubleshooting, System File Checker, storage management and Event Viewer.

The project helped me develop practical experience in monitoring systems, checking network connectivity, troubleshooting Windows system files and investigating system events. I also documented each task and included screenshots as evidence of the work completed.

Overall, this project demonstrates my understanding of basic IT Support troubleshooting processes and my ability to document technical work.